

Correlation and Covariance

Question no 1:- Define Covariance and explain how it differs from Correlation in terms of scale and interpretation.

Answer:- Covariance

- Definition: Covariance measures how two variables change together.
 - Positive covariance → when one variable increases, the other tends to increase.
 - Negative covariance → when one increases, the other tends to decrease.
- Formula:

$$\text{Cov}(X,Y) = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

- Scale: Covariance is expressed in the combined units of the two variables (e.g., kg·cm, dollars·hours).
- Interpretation: Only tells the direction of the relationship (positive/negative), not its strength.

Correlation

- Definition: Correlation standardizes covariance to measure both direction and strength of the relationship.
- Formula:

$$r = \frac{\text{Cov}(X,Y)}{\sigma_X \cdot \sigma_Y}$$

- Scale: Correlation is unitless, always between -1 and +1.
- Interpretation:
 - $r=+1$: perfect positive relationship
 - $r=-1$: perfect negative relationship
 - $r=0$: no linear relationship

Key Difference

- **Covariance**: raw measure, depends on scale of variables, hard to compare across datasets.
- **Correlation**: standardized measure, easy to interpret, universally comparable.

Example (Retail Context)

- If we compute covariance between **Units Sold** and **Revenue**, we might get a large positive number (say 12,000). That tells us they move together, but the magnitude depends on units.

Correlation and Covariance

- If we compute correlation, we might get $r=0.92$. That tells us the relationship is **strong and positive**, regardless of scale.

In short:

- Covariance = direction only, scale-dependent
- Correlation = direction + strength, scale-free

Question no 2:- What does a positive, negative, and zero covariance indicate about the relationship between two variables?

Answer:- Interpret covariance values in terms of the relationship between two variables:

♦ Positive Covariance

- Indicates that the two variables move in the same direction.
- When one increases, the other tends to increase as well.
- Example: In retail, if Units Sold and Revenue have positive covariance, higher sales volumes are associated with higher revenue.

♦ Negative Covariance

- Indicates that the two variables move in opposite directions.
- When one increases, the other tends to decrease.
- Example: If Product Price and Units Sold have negative covariance, higher prices are linked to lower sales volumes.

Zero Covariance

- Suggests **no linear relationship** between the two variables.
- Changes in one variable do not predict changes in the other.
- Example: If **Employee Age** and **Daily Sales** have zero covariance, age does not influence sales performance.

Key Point

Covariance tells you the **direction** of the relationship but not its **strength**. For strength and comparability, you'd use **correlation**, which standardizes covariance into a unit-free measure between -1 and $+1$.

Question no 3:- Discuss the limitations of covariance as a measure of relationship between two variables. Why is correlation preferred in many cases?

Correlation and Covariance

Answer:- Covariance is useful, but it has some important limitations that explain why correlation is often preferred.

Limitations of Covariance

1. Scale-dependent
 - Covariance values depend on the units of measurement.
 - Example: If sales are measured in dollars and advertising spend in thousands of dollars, the covariance will change if you switch units (e.g., dollars vs. rupees).
2. Unbounded values
 - Covariance can take any value (positive, negative, large, small).
 - This makes it hard to interpret or compare across datasets.
3. Direction only, not strength
 - A positive covariance tells you variables move together, but doesn't tell you how strongly.
 - Two datasets can have the same sign of covariance but very different levels of association.
4. Not standardized
 - Because it's not normalized, covariance cannot be directly compared between different variable pairs.

Correlation is Preferred

- **Standardized measure:** Correlation divides covariance by the product of standard deviations, removing scale effects.
- **Bounded range:** Always between -1 and $+1$, making interpretation straightforward.
- **Strength + direction:** Correlation shows both how strong and in which direction the relationship lies.
- **Comparability:** Correlation values can be compared across different datasets, variables, or contexts.

Example (Retail Context)

- Covariance between **Units Sold** and **Revenue** might be **12,000**. That's positive, but the magnitude is hard to interpret.
- Correlation might be **0.92**, which clearly indicates a **strong positive relationship**.

In short:

- **Covariance** tells you *direction only* and is scale-dependent.

Correlation and Covariance

- **Correlation** tells you *direction + strength* and is scale-free, which is why it's the go-to measure in most analyses.

Question no 4:- Explain the difference between Pearson's correlation coefficient and Spearman's rank correlation coefficient. When would you prefer to use Spearman's correlation?

Answer:- A clear breakdown of the two correlation measures:

Pearson's Correlation Coefficient (r)

- **Definition:** Measures the **linear relationship** between two continuous variables.
- **Assumptions:**
 - Variables are normally distributed.
 - Relationship is linear.
 - Sensitive to outliers.
- **Scale:** Ranges from -1 to $+1$.
- **Interpretation:**
 - $r=+1$: perfect positive linear relationship
 - $r=-1$: perfect negative linear relationship
 - $r=0$: no linear relationship

Spearman's Rank Correlation Coefficient (ρ)

- **Definition:** Measures the **monotonic relationship** between two variables using their ranks rather than raw values.
- **Assumptions:**
 - Does not require normal distribution.
 - Works with ordinal data or non-linear monotonic relationships.
 - Less sensitive to outliers.
- **Scale:** Also ranges from -1 to $+1$.
- **Interpretation:** Same as Pearson, but based on ranked data.

Key Differences

Feature	Pearson's r	Spearman's ρ
Data type	Continuous, interval/ratio	Ordinal or ranked
Relationship measured	Linear	Monotonic (increasing/decreasing)
Sensitivity to outliers	High	Lower
Assumptions	Normality, linearity	None (non-parametric)

Correlation and Covariance

Prefer Spearman's

- When data are **ordinal** (e.g., customer satisfaction ratings: poor, fair, good, excellent).
- When the relationship is **non-linear but monotonic** (e.g., income vs. happiness: happiness rises with income but levels off).
- When data contain **outliers** that could distort Pearson's correlation.

In short:

- **Pearson's r** → best for linear, continuous, normally distributed data.
- **Spearman's ρ** → best for ranked, ordinal, or non-linear monotonic relationships.

Question no 5:- If the correlation coefficient between two variables X and Y is 0.85, interpret this value in context. Can you infer causation from this value? Why or why not?

Answer:- A correlation coefficient of 0.85 between two variables X and Y indicates a strong positive linear relationship.

Interpretation

- As X increases, Y tends to increase as well.
- The strength is high (close to +1), meaning the data points lie close to a straight upward-sloping line.
- Example: If X = advertising spend and Y = sales revenue, then higher spending is strongly associated with higher revenue.

Infer Causation?

No. Correlation does **not imply causation**.

- Correlation only shows that two variables move together.
- It does not prove that changes in X cause changes in Y.
- There may be:
 - **Confounding variables** (e.g., seasonality affecting both advertising and sales).
 - **Reverse causation** (e.g., higher sales leading to more advertising).
 - **Coincidental correlation** (spurious relationships).

Key Point

- **Correlation (0.85)** → strong positive association.
- **Causation** → requires controlled experiments or deeper statistical methods (like regression with confounder control, randomized trials).

Question no 6:- Using the dataset below, calculate the covariance between X and Y.

Correlation and Covariance

X	2	4	6	8
Y	3	7	5	10

Answer:- Dataset

- $X=[2,4,6,8]$
- $Y=[3,7,5,10]$
- $n=4$

Step 1: Means

$$X^- = \frac{2+4+6+8}{4} = \frac{20}{4} = 5$$

$$Y^- = \frac{3+7+5+10}{4} = \frac{25}{4} = 6.25$$

Step 2: Deviations

X	Y	$X-X^-$	$Y-Y^-$	Product
2	3	-3	-3.25	9.75
4	7	-1	0.75	-0.75
6	5	+1	-1.25	-1.25
8	10	+3	+3.75	11.25

Sum of products = $9.75-0.75-1.25+11.25=19.0$

Step 3: Covariance Formula

$$\text{Cov}(X,Y) = \frac{\sum(X_i-X^-)(Y_i-Y^-)}{N-1}$$

$$\text{Cov}(X,Y) = \frac{19.0}{3} \approx 6.33$$

The **covariance between X and Y is ≈ 6.33 .**

Interpretation

Correlation and Covariance

- Since it's **positive**, X and Y tend to move in the **same direction**: when X increases, Y generally increases too.
- The magnitude (6.33) depends on the scale of X and Y, so it's not directly comparable across datasets. For strength, you'd compute the **correlation coefficient**.

Question no 7:- Compute the Pearson correlation coefficient between variables A and B:

A	10	20	30	40	50
B	8	14	18	24	28

Answer:- Dataset

- A=[10,20,30,40,50]
- B=[8,14,18,24,28]
- n=5

Step 1: Means

$$A^- = \frac{10+20+30+40+50}{5} = \frac{150}{5} = 30$$
$$B^- = \frac{8+14+18+24+28}{5} = \frac{92}{5} = 18.4$$

Step 2: Deviations and Products

A	B	A-A ⁻	B-B ⁻	Product
10	8	-20	-10.4	208.0
20	14	-10	-4.4	44.0
30	18	0	-0.4	0.0
40	24	+10	+5.6	56.0
50	28	+20	+9.6	192.0

- Sum of products = **500.0**

Step 3: Covariance

Correlation and Covariance

$$\text{Cov}(A,B) = \frac{\sum(A_i - \bar{A})(B_i - \bar{B})}{n-1}$$

$$\text{Cov}(A,B) = \frac{500}{4} = 125.0$$

Step 4: Standard Deviations

$$s_A^2 = \frac{\sum(A_i - \bar{A})^2}{n-1} = \frac{1000}{4} = 250 \Rightarrow S_A = \sqrt{250} \approx 15.81$$

$$s_B^2 = \frac{\sum(B_i - \bar{B})^2}{n-1} = \frac{229.2}{4} = 57.3 \Rightarrow S_B = \sqrt{57.3} \approx 7.57$$

Step 5: Pearson Correlation

$$r = \frac{\text{Cov}(A,B)}{S_A \cdot S_B}$$

$$r = \frac{125}{15.81 \cdot 7.57} = \frac{125}{119.7} \approx 0.95$$

The **Pearson correlation coefficient** between A and B is ≈ 0.95 .

Interpretation

- This indicates a **very strong positive linear relationship** between A and B.
- As A increases, B increases almost proportionally.

Question no 8:- The following table shows heights (in cm) and weights (in kg) of 5 students. Find the correlation coefficient between Height and Weight.

Height	150	160	165	170	180
Weight	50	55	58	62	70

Answer:- Dataset

- Heights (cm): [150, 160, 165, 170, 180]
- Weights (kg): [50, 55, 58, 62, 70]
- n=5

Step 1: Means

Correlation and Covariance

$$\begin{aligned} H^- &= \frac{150+160+165+170+180}{5} = \frac{825}{5} = 165 \\ W^- &= \frac{50+55+58+62+70}{5} = \frac{295}{5} = 59 \end{aligned}$$

Step 2: Deviations and Products

Height	Weight	H-H ⁻	W-W ⁻	Product
150	50	-15	-9	135
160	55	-5	-4	20
165	58	0	-1	0
170	62	+5	+3	15
180	70	+15	+11	165

- Sum of products = **335**

Step 3: Covariance

$$\begin{aligned} \text{Cov}(H,W) &= \frac{\sum(H_i - H^-)(W_i - W^-)}{n-1} \\ \text{Cov}(H,W) &= \frac{335}{4} = 83.75 \end{aligned}$$

Step 4: Standard Deviations

$$\begin{aligned} \text{For Height:} \\ s_H^2 &= \frac{(-15)^2 + (-5)^2 + 0^2 + 5^2 + 15^2}{4} = \frac{500}{4} = 125 \Rightarrow S_H = \sqrt{125} \approx 11.18 \\ \text{For Weight:} \\ s_W^2 &= \frac{(-9)^2 + (-4)^2 + (-1)^2 + 3^2 + 11^2}{4} = \frac{252}{4} = 63 \Rightarrow S_W = \sqrt{63} \approx 7.94 \end{aligned}$$

Step 5: Pearson Correlation

Correlation and Covariance

$$r = \frac{\text{Cov}(H,W)}{S_H \cdot S_W}$$
$$r = \frac{83.75}{11.18 \cdot 7.94} = \frac{83.75}{88.8} \approx 0.94$$

The correlation coefficient between Height and Weight is ≈ 0.94 .

Interpretation

- This indicates a very strong positive linear relationship.
- Taller students tend to weigh more, and the relationship is almost linear.

Question no 9:- Given the dataset below, determine whether there is a positive or negative correlation between X and Y. (No need for exact calculation, just reasoning.)

X	1	2	3	4	5
Y	15	12	9	7	3

Answer:- Dataset:

- $X=[1,2,3,4,5]$
- $Y=[15,12,9,7,3]$

Reasoning

- As X increases, Y decreases.
- This consistent opposite movement indicates a negative relationship.
- If we were to calculate correlation, the coefficient would be negative (close to -1 , since the pattern is nearly linear).

Conclusion

There is a negative correlation between X and Y: higher values of X are associated with lower values of Y.

Question no 10:- Two investment portfolios have the following returns (%) over 5 years. Compute the covariance and correlation coefficient, and interpret whether the portfolios move together.

Correlation and Covariance

Year	Portfolio A	Portfolio B
1	8	6
2	10	9
3	12	11
4	9	8
5	11	10

Answer:- compute both covariance and correlation coefficient for the two portfolios.

Dataset:-

Year	Portfolio A	Portfolio B
1	8	6
2	10	9
3	12	11
4	9	8
5	11	10

n=5

Step 1: Means

$$A^- = \frac{8+10+12+9+11}{5} = \frac{50}{5} = 10$$

$$B^- = \frac{6+9+11+8+10}{5} = \frac{44}{5} = 8.8$$

Step 2: Deviations and Products

A	B	A-A ⁻	B-B ⁻	Product
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Correlation and Covariance

8	6	-2	-2.8	5.6
10	9	0	0.2	0.0
12	11	+2	+2.2	4.4
9	8	-1	-0.8	0.8
11	10	+1	+1.2	1.2

- Sum of products = **12.0**

Step 3: Covariance

$$\text{Cov}(A,B) = \sum (A_i - \bar{A})(B_i - \bar{B}) / n - 1$$

$$\text{Cov}(A,B) = 12.04 = 3.0$$

Step 4: Standard Deviations

- For A:

$$s_A^2 = (-2)^2 + 0^2 + 2^2 + (-1)^2 + 1^2 = 10 = 2.5$$

$$s_A = 2.5 \approx 1.58$$

- For B:

$$s_B^2 = (-2.8)^2 + 0.2^2 + 2.2^2 + (-0.8)^2 + 1.2^2 = 14.84 = 3.7$$

$$s_B = 3.7 \approx 1.92$$

Step 5: Correlation Coefficient

$$r = \text{Cov}(A,B) / s_A \cdot s_B$$

$$r = 3.0 / (1.58 \cdot 1.92) = 3.0 / 3.04 \approx 0.99$$

- Covariance = 3.0
- Correlation coefficient (r) $\approx$ 0.99

Interpretation

- The covariance is positive, meaning the portfolios tend to move in the same direction.

Correlation and Covariance

- The correlation is extremely close to +1, indicating a very strong positive linear relationship.
 - In practice: when Portfolio A's returns rise, Portfolio B's returns almost always rise proportionally, and vice versa.
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